

Physics-Informed Neural Networks for FitzHugh–Nagumo Neuron Modeling

Project Notes and Technical Report

Introduction

This project investigates the application of Physics-Informed Neural Networks (PINNs) to neuron modeling using the FitzHugh–Nagumo model.

In many biological and neural systems, only a subset of the system variables can be measured directly. Important internal states often remain hidden due to experimental limitations. Traditional neural networks learn patterns solely from available observations, whereas PINNs combine observational data with known governing equations.

The primary objective of this study is to determine whether incorporating neuron dynamics into the learning process improves reconstruction of hidden neural states compared to a standard Neural Network.

The FitzHugh–Nagumo model was selected because it is a well-established mathematical representation of neuronal excitation and recovery dynamics.

The central research question addressed in this study is:

Can a Physics-Informed Neural Network reconstruct hidden neural states more accurately than a standard Neural Network when only sparse observations are available?

Project Resources

Google Colab Notebook

The complete implementation of the Physics-Informed Neural Network for FitzHugh–Nagumo neuron modeling can be accessed through the following Google Colab notebook:

https://colab.research.google.com/drive/1QNApE1JDZItIAJZbBR_rjGVq-PfT0P0f?usp=sharing

The notebook contains:

- Numerical simulation of the FitzHugh–Nagumo equations
- Synthetic dataset generation
- Standard Neural Network implementation
- Physics-Informed Neural Network implementation
- Automatic differentiation using PyTorch
- Training and evaluation routines
- Visualization and comparison plots

Model Information Source

Additional information regarding the FitzHugh–Nagumo neuron model can be found at:

https://en.wikipedia.org/wiki/FitzHugh%E2%80%93Nagumo_model

FitzHugh–Nagumo Model

The FitzHugh–Nagumo model is a simplified version of more complex neuron models such as the Hodgkin–Huxley model.

Despite its simplicity, it captures two essential aspects of neuronal behavior:

- Excitation
- Recovery

The model consists of two coupled differential equations.

The first variable, v , represents the membrane potential and describes the electrical activity of the neuron.

The second variable, w , represents the recovery process that gradually returns the neuron to its resting state after excitation.

The model parameters a , b , and τ control the behavior of the system, while I represents an externally applied current.

Because the two variables are dynamically coupled, information about one variable contains information about the other. This property makes the model ideal for evaluating hidden-state reconstruction using PINNs.

Dataset Generation

Experimental recordings were not used in this study.

Instead, synthetic data was generated by numerically solving the FitzHugh–Nagumo differential equations using SciPy’s ODE solver.

The simulation produced two signals:

- Membrane potential $v(t)$
- Recovery variable $w(t)$

Since the signals are generated directly from the governing equations, they serve as ground-truth data.

This enables accurate evaluation of model performance because the exact values of all variables are known.

Synthetic datasets are particularly useful for validating PINN methodologies because the underlying physical model is fully known.

Research Question

The primary objective of this study is to evaluate whether a Physics-Informed Neural Network can recover hidden neural states more accurately than a conventional Neural Network.

To create a realistic hidden-state reconstruction problem, the recovery variable $w(t)$ was intentionally removed from the training data.

Only sparse observations of the membrane potential $v(t)$ were provided during training.

The recovery variable was never directly observed by either model.

This setup resembles practical neuroscience scenarios in which internal physiological variables cannot be measured directly.

The challenge for both models is to reconstruct the hidden recovery variable solely from the observed membrane potential.

Standard Neural Network

The first model is a fully connected feedforward Neural Network.

The input to the network is time:

$$t$$

The network predicts both neuron states:

$$\hat{v}(t), \quad \hat{w}(t)$$

The loss function only compares the predicted membrane potential with the observed membrane potential.

Since the true recovery variable is hidden, no supervision is provided for w .

Consequently, the model relies entirely on patterns present in the observed data.

This represents a purely data-driven learning approach.

Physics-Informed Neural Network

The Physics-Informed Neural Network uses the same architecture as the standard Neural Network.

The key difference is the addition of physics-based constraints.

The loss function contains two components:

1. Data Loss
2. Physics Loss

The data loss measures agreement between predicted and observed membrane potentials.

The physics loss is computed using the FitzHugh–Nagumo differential equations.

Automatic differentiation is used to compute temporal derivatives directly from the network outputs.

Residuals of the governing equations are then minimized during training.

As a result, the PINN learns from both observational data and scientific knowledge encoded in the neuron dynamics.

Why PINNs Improve Hidden-State Reconstruction

A standard Neural Network learns exclusively from observed data.

When important variables are hidden, the network receives very little information about their behavior.

Physics-Informed Neural Networks address this limitation by incorporating prior scientific knowledge.

The FitzHugh–Nagumo equations establish a mathematical relationship between membrane potential and recovery dynamics.

Therefore, even when the recovery variable is not observed, the governing equations provide indirect supervision.

The physical laws effectively serve as an additional source of information during training.

This additional guidance enables more accurate reconstruction of hidden neural states.

Results

The hidden recovery variable predictions were compared against the ground-truth solution obtained from numerical simulation.

The standard Neural Network showed noticeable deviations from the true recovery dynamics.

In contrast, the Physics-Informed Neural Network closely followed the true trajectory of the hidden state.

This result is particularly significant because the PINN never observed the true recovery variable during training.

The improvement demonstrates the effectiveness of incorporating physical knowledge into the learning process.

Quantitative Results

Model performance was evaluated using Root Mean Square Error (RMSE).

Membrane Potential

- Standard Neural Network RMSE = 0.193
- Physics-Informed Neural Network RMSE = 0.170

Recovery Variable

- Standard Neural Network RMSE = 0.156
- Physics-Informed Neural Network RMSE = 0.099

The improvement is particularly pronounced for the hidden recovery variable.

This demonstrates that the governing differential equations provide valuable information that improves hidden-state estimation.

Important Mathematical Equations

FitzHugh–Nagumo Equations

The neuron dynamics are governed by:

$$\frac{dv}{dt} = v - \frac{v^3}{3} - w + I \quad (1)$$

$$\tau \frac{dw}{dt} = v + a - bw \quad (2)$$

or equivalently

$$\frac{dw}{dt} = \frac{v + a - bw}{\tau} \quad (3)$$

Variable Definitions

- v = membrane potential
- w = recovery variable
- I = external input current
- a, b, τ = model parameters

Neural Network Prediction

The network predicts

$$NN(t) = \begin{bmatrix} \hat{v}(t) \\ \hat{w}(t) \end{bmatrix} \quad (4)$$

Data Loss

$$L_{data} = \frac{1}{N} \sum_{i=1}^N \left(v_i^{obs} - \hat{v}_i \right)^2 \quad (5)$$

Physics Residuals

$$f_v = \frac{d\hat{v}}{dt} - \left(\hat{v} - \frac{\hat{v}^3}{3} - \hat{w} + I \right) \quad (6)$$

$$f_w = \frac{d\hat{w}}{dt} - \frac{\hat{v} + a - b\hat{w}}{\tau} \quad (7)$$

For a perfect solution:

$$f_v = 0$$

$$f_w = 0$$

Physics Loss

$$L_{physics} = \frac{1}{N} \sum_{i=1}^N (f_{v,i}^2 + f_{w,i}^2) \quad (8)$$

Total PINN Loss

$$L_{total} = L_{data} + \lambda L_{physics} \quad (9)$$

For this project:

$$\lambda = 0.1$$

Root Mean Square Error

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2} \quad (10)$$

Advantages of PINNs

Physics-Informed Neural Networks provide several advantages:

- Ability to learn with sparse data
- Incorporation of scientific knowledge
- Improved hidden-state reconstruction
- Better generalization
- Physically consistent predictions

Conclusion

This study investigated the use of Physics-Informed Neural Networks for FitzHugh–Nagumo neuron modeling.

A hidden-state reconstruction problem was created by withholding the recovery variable during training and providing only sparse observations of the membrane potential.

The Physics-Informed Neural Network incorporated neuron dynamics through the FitzHugh–Nagumo differential equations, enabling the model to learn from both data and governing physics.

Experimental results demonstrated that the PINN achieved lower prediction errors than a standard Neural Network, particularly for the hidden recovery variable.

These findings highlight the value of integrating scientific knowledge into machine learning models, especially when measurements are incomplete or data is limited.

Physics-Informed Neural Networks represent a promising methodology for neuroscience, neural engineering, biomedical modeling, and scientific machine learning applications.